

SMILE: A Hybrid Memory-Driven Dialogue Manager for a Social Companion Robot

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December 9, 2025

Abstract

This report presents SMILE (Social Memory Integrated Learning Environment), a hybrid memory-augmented dialogue system designed for embodied social companion robots. Unlike purely end-to-end LLM-based conversational agents, SMILE adopts a controlled architecture where memory retrieval, dialogue state management, and behavioral constraints are implemented deterministically through code, while the LLM serves exclusively as a linguistic realizer. The system integrates real-time face recognition for persistent user identification, offline speech processing for privacy-preserving interaction, and a three-layer memory structure comprising working memory (recent conversational turns), semantic memory (stable user attributes), and episodic memory (discrete past experiences). This architecture addresses critical limitations of stateless LLMs in assistive robotics contexts, where predictability, transparency, and protection against hallucination are paramount. We describe SMILE’s modular implementation, including its finite-state dialogue manager, LLM-based memory consolidation pipeline, and deterministic retrieval mechanisms that balance personalization with privacy. The report analyzes fundamental trade-offs between cognitive plausibility and computational tractability in memory retrieval, examines the limitations of prompt-based behavioral control, and proposes a two-phase evaluation protocol designed to validate technical robustness and user acceptance before deployment with vulnerable populations. We critically assess challenges including ASR noise propagation, semantic-episodic memory boundaries, and the tension between rule-based retrieval (ensuring safety) and flexible associative recall (enabling natural conversation). The work concludes by situating SMILE within the broader landscape of memory-augmented conversational AI and cognitive robotics, identifying open research questions at the intersection of human memory models and artificial companion systems, and outlining the path from research prototype to clinically validated assistive technology for older adults and individuals with cognitive impairments.

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1 Introduction

Social robots are progressively moving from controlled laboratory environments to real-world settings such as elderly care, cognitive rehabilitation, education, and everyday assistance. In these contexts, robots are not merely expected to execute isolated tasks, but to sustain long-term, meaningful relationships with users. This notion of “companionship” involves continuity over time, personalization, and the ability to remember what has happened in previous encounters. Despite significant progress in perception, navigation and speech technologies, most deployed social robots still operate with very limited memory. They can recognise a face, transcribe speech, or answer questions, but they rarely *remember* who they are talking to in a structured way, nor can they reliably relate current events to past experiences with the same person. As a consequence, interactions often feel shallow and repetitive, and users may perceive the robot as less intelligent or less socially credible than its physical embodiment suggests. Large Language Models (LLMs) have recently transformed the landscape of natural language interaction. Their ability to generate fluent, contextually appropriate responses makes them attractive candidates for powering the dialogue capabilities of social robots. However, LLMs are, by design, stateless: they only “remember” what is explicitly provided in the input prompt or encoded in their parameters. This creates a tension between the apparent conversational intelligence of the model and the lack of persistent, structured memory required for long-term social interaction. In this work we present SMILE (Social Memory-Integrated Language Environment), a memory-augmented dialogue system designed for social companion robots. SMILE combines real-time face recognition, continuous spoken dialogue, and a hybrid memory system that maintains:

- a *short-term working memory* of recent utterances,
- a *long-term semantic memory* with stable user attributes (name, estimated age, interests, goals, personality traits),
- a *growing episodic memory* of concrete experiences and events mentioned by the user.

The system is implemented as a local, fully controllable pipeline that integrates automatic speech recognition (ASR), text-to-speech (TTS), face-based user identification, and an LLM hosted via Ollama. The aim is not only to sustain more coherent conversations across turns, but also to preserve a personalised history that can be revisited and analysed for research or clinical purposes, for example in contexts involving older adults or individuals with cognitive impairments.

1.1 Problem Definition and Research Goals

Building a social robot capable of long-term companionship poses several interrelated challenges. In the context of SMILE, we address four core problems:

1. **Persistent identity recognition.** The robot must be able to recognise a person across multiple encounters, associate them with a stable internal representation, and retrieve their past information whenever they reappear. In SMILE, this is achieved via face recognition using FaceNet embeddings and JSON-based profile storage. Persistent identity forms the foundation for all subsequent memory operations.
2. **Multi-layer memory representation.** Human memory is not monolithic; it comprises multiple interacting systems including working memory, semantic memory and episodic memory. For a social robot, an analogous structure is needed: working memory for the current conversation, semantic memory describing stable user attributes, and episodic memory capturing past experiences. Designing such a representation requires careful decisions about what to store, in which format, and through which update mechanisms.
3. **Contextually appropriate memory use.** Having access to stored information is insufficient; the robot must know when and how to use it. Indiscriminately injecting all

available memory into an LLM prompt can lead to hallucinations, inappropriate associations, or irrelevant references to the past. SMILE must selectively combine working memory, episodic memories and profile information based on the current conversational context and dialogue state.

4. **Automatic memory consolidation.** At the end of each session, the robot should “consolidate” the encounter into persistent memory. In SMILE, this is accomplished by prompting the LLM to summarise the recent dialogue and extract structured episodes, including titles, timeframes, descriptions, and retrieval tags. The user profile is then updated with new or refined information about interests, goals and personality traits. This post-hoc consolidation step enables longitudinal memory development without manual annotation.

1.2 Contributions and Report Structure

This report makes the following contributions:

- **Hybrid memory architecture:** We present a cognitively-inspired memory system that separates working, semantic, and episodic memory with distinct access patterns and persistence characteristics, adapted to the constraints of LLM-based dialogue systems.
- **Deterministic dialogue control:** We demonstrate how explicit finite-state machine dialogue management combined with structured prompt engineering can achieve predictable, safe behavior while leveraging LLM linguistic capabilities.
- **Automatic profile evolution:** We implement LLM-based memory consolidation that extracts structured semantic attributes and episodic events from natural conversation, enabling user profiles to develop organically over multiple sessions.
- **Feasible evaluation protocol:** We propose a two-phase evaluation protocol that balances methodological rigor with feasibility, providing a pathway from technical validation to longitudinal deployment studies.

The remainder of this report is structured as follows. Section 2 reviews relevant literature on memory systems in human cognition, cognitive robotics, and LLM-based agents, positioning SMILE within this landscape. Section 3 provides a high-level overview of the complete system architecture, describing how perception, dialogue management, and memory integrate into a coherent pipeline. Section 4 details the design and implementation of SMILE’s three-layer memory system, including the principles governing memory access and consolidation. Section 5 examines the finite-state machine that controls conversational flow and mediates memory retrieval. Section 6 describes technical implementation details including face recognition, speech processing, LLM integration, and concurrency management. Section 7 proposes a practical evaluation protocol for assessing memory accuracy, dialogue coherence, and user acceptance. Section 8 critically examines the system’s limitations, discusses architectural trade-offs, and identifies open research questions. Finally, Section 9 summarizes key findings and outlines directions for future work. Through this work, we aim to demonstrate that a carefully designed hybrid architecture—combining deterministic memory control with LLM linguistic generation—can provide a practical foundation for memory-augmented social robotics while maintaining the transparency and predictability essential for assistive applications with vulnerable populations.

2 Background and Related Work

The development of conversational robots has historically intersected with three distinct research domains: cognitive robotics, memory modelling, and dialogue systems. The SMILE framework lies at the intersection of all three, combining classical insights from artificial memory architectures with modern Large Language Model (LLM) capabilities. This section provides an overview of these areas, highlighting similarities and discrepancies with existing approaches, and clarifying the conceptual foundations upon which SMILE is built.

2.1 Memory Systems in Human Cognition

Understanding artificial memory systems for robots requires first examining how memory is structured in biological systems. Research in cognitive psychology has identified multiple memory subsystems through studies of amnesic patients and neurological observations throughout the second half of the 20th century [Conway and Pleydell-Pearce, 2001]. Based on these findings, Atkinson and Shiffrin proposed the Multi-Store Model [Atkinson and Shiffrin, 1968], which distinguishes three fundamental memory types: sensory memory, which briefly processes perceptual information; short-term (or working) memory, which maintains information through active rehearsal; and long-term memory, which provides persistent storage of knowledge and experiences. Figure 1 illustrates this taxonomy, showing how information flows from sensory input through working memory to long-term storage. Working memory holds information for limited durations (typically 15–25 seconds) and has a constrained capacity of approximately 7 ± 2 “chunks” of information [Miller, 1956, Cowan, 2001]. Long-term memory subdivides into declarative memory (explicit knowledge of facts and events) and procedural memory (implicit knowledge of skills and behaviors). Declarative memory further separates into episodic memory—which stores personally experienced events with spatial and temporal context—and semantic memory, which contains general factual knowledge. As Conway and Pleydell-Pearce emphasize [Conway and Pleydell-Pearce, 2001], episodic and semantic memory are deeply interlinked: new semantic knowledge often emerges from the consolidation of episodic experience. This observation is particularly relevant for social robots that must build understanding through interaction. Moreover, the temporal nature of episodic memory is crucial: it manages not just what happened, but when and where it occurred, providing the contextual grounding necessary for coherent long-term interaction.

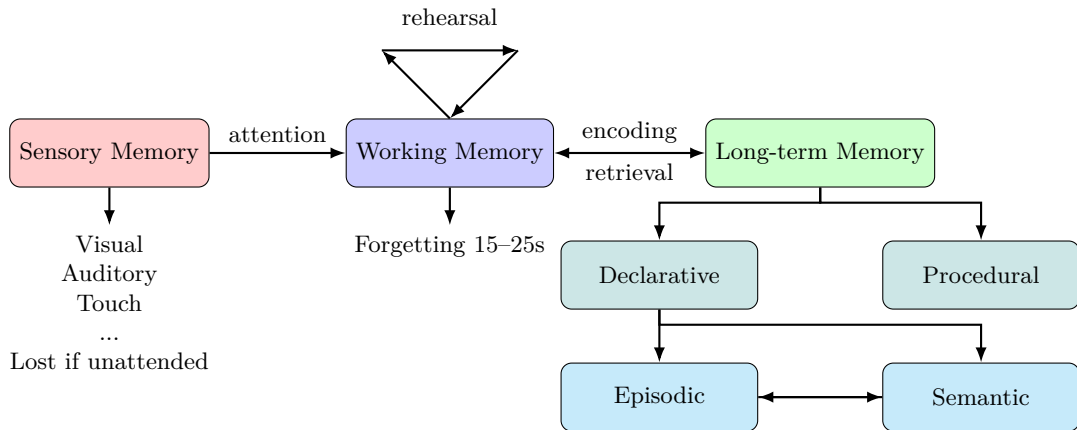


Figure 1: Taxonomy of human memory systems, illustrating the flow from sensory input to working memory and the division of long-term memory into declarative (episodic and semantic) and procedural components.

2.2 Memory in Cognitive Robotics

Memory has long been recognized as a foundational component of cognitive robotics architectures. Traditional systems view memory as a layered structure inspired by human cognition, incorporating procedural memory for motor behaviors, semantic memory for stable knowledge, and episodic memory for past interactions and experiences [Laird et al., 1987, Laird and Rosenbloom, 2014]. Peller-Konrad et al. argue [Peller-Konrad et al., 2023] that memory serves as “a central active component” of cognitive architectures, mediating between semantic and sensorimotor representations while orchestrating data flow between processes. This active role distinguishes modern memory systems from passive storage: memory must adapt to incoming data, learn representations, and provide services for reasoning, prediction, and action parametrization. A key insight from cognitive robotics is that memory is not used continuously but acts as a contextual trigger: it is selectively accessed in response to cues, relevance, or time-dependent mechanisms [Tulving, 2002]. This reflects observations in human cognition, where episodic recall is reactive rather than constantly active. In practice, this means memory systems must support efficient retrieval based on temporal, spatial, or semantic queries rather than maintaining all past information in active working memory.

2.2.1 Episodic Memory Architectures

Several cognitive architectures have implemented episodic memory with varying degrees of sophistication. Soar [Laird et al., 1987, Laird, 2012] includes an episodic memory that stores snapshots of working memory states, enabling the system to recall past experiences and ground decision-making in prior contexts. The ISAC architecture [Kawamura et al., 2008] stores episodic knowledge as abstractions of sensory representations enriched with goals, actions, outcomes, and valuations, organised in multiple layers to support efficient retrieval. CRAM [Beetz et al., 2010] represents episodes as “narrative-enabled episodic memories” (NEEMs)—first-order time-interval logic expressions enriched with detailed perceptual and procedural events. This representation enables reasoning about complete action traces and supports explanation of system behaviour. Across these systems, common principles emerge regarding the design of robotic episodic memory. As noted by Peller-Konrad et al. [Peller-Konrad et al., 2023], effective memory architectures for humanoid robots typically require active processing, multi-modal and inherently temporal representations, associative organisation, and introspectability. These principles are reflected to varying degrees in existing architectures such as Soar, ISAC, CRAM, and ArmarX. Together, these approaches illustrate how structured episodic memory can support complex sensorimotor integration, reasoning over past experiences, and explainability of robot behaviour in real-world tasks.

2.3 LLM-Based Dialogue in Social Robots

Integrating Large Language Models into embodied social robots is an active and rapidly evolving research area. Irfan et al. [Irfan et al., 2025] examine the deployment of LLM-driven dialogue in companion robots for older adults, highlighting both the promise and the persistent challenges of this approach. While LLMs excel at generating fluent and contextually appropriate utterances, they suffer from several well-known limitations: character inconsistency, lack of persistent user modelling, and the inability to maintain continuity across sessions. Existing systems typically mitigate these issues through “persona prompts” (system-level messages prescribing identity and behaviour) or by retrieving snippets of past dialogue. However, as shown by Irfan et al., these strategies scale poorly over extended deployments and may yield contradictions or inappropriate callbacks. Their empirical results indicate that older adults quickly notice such inconsistencies and express frustration when robots fail to recall prior conversations or personal details. This line of work underscores a broader theme in social robotics: the need for structured and reliable memory mechanisms to support rapport, trust and long-term engagement. As emphasised

by Peller-Konrad et al. [Peller-Konrad et al., 2023], robots interacting with older adults or individuals with cognitive impairments benefit significantly from the ability to recall past events, preferences, routines and emotional states. These observations motivate the development of memory architectures specifically designed for assistive social robots operating over extended periods.

2.3.1 Memory in Large Language Models

Large Language Models fundamentally lack memory in the traditional cognitive sense: their knowledge is static and encoded in model parameters, and they do not retain information across interactions unless it is explicitly reintroduced in the prompt. As detailed in the comprehensive survey by Zhang et al. [Zhang et al., 2025], implementing memory for LLM-based agents requires external mechanisms. The authors identify three categories of artificial memory closely aligned with classical cognitive taxonomy:

1. **Working memory:** short-term conversational context maintained within the prompt window.
2. **Semantic memory:** stable facts or attributes stored externally and injected into the prompt when relevant.
3. **Episodic memory:** collections of past interactions, typically stored as logs or summaries and retrieved on demand.

Because LLMs possess no intrinsic persistent memory, long-term coherence must be achieved through explicit mechanisms such as context management (selectively providing relevant prior turns), Retrieval-Augmented Generation (RAG), external memory modules interfaced with the LLM, or summarisation pipelines that distil conversational history into stable user profiles. These approaches vary in complexity and reliability, but they share the same underlying objective: compensating for the LLM’s inherent statelessness in order to support sustained, personalised interaction.

2.4 Gap Analysis and SMILE’s Positioning

The literature reviewed above highlights several limitations that constrain current approaches to memory in social robots and LLM-based dialogue systems. Most LLM-driven conversational agents rely heavily on working-context windows, resulting in fragmented or unstable long-term coherence, while current robot–LLM integrations rarely maintain evolving user models across sessions, a limitation noted in recent evaluations of conversational robots in assistive settings [Irfan et al., 2025]. At the same time, established cognitive robotics architectures such as Soar or CRAM offer rich episodic and semantic memory mechanisms, yet their symbolic complexity and computational overhead make them difficult to integrate with real-time LLM dialogue on embedded or domestic platforms. Furthermore, research in socially assistive robotics highlights that continuity, predictable behaviour, and validation of personal identity are particularly critical when interacting with older adults or vulnerable users [Robinson et al., 2013, Abdi et al., 2018], yet few existing systems provide lightweight and transparent memory solutions tailored to these needs. SMILE is positioned precisely within this gap. The system adopts a hybrid design that preserves the strengths of LLM-based natural language generation while introducing explicit mechanisms for identity continuity and longitudinal personalisation. Face recognition enables persistent user identification across sessions, directly linking perception with memory. Conversational episodes are logged as human-readable JSON files, forming an explicit episodic trace that supports both inspection and reproducibility. After each session, LLM-based summarisation extracts salient user information to update a long-term semantic profile, allowing the system to refine

its understanding of individual users over time without relying on heavy symbolic inference. Crucially, memory access in SMILE is deterministic: identity is established before dialogue begins, and all retrieval operations operate over structured data rather than uncontrolled model recall. This enhances predictability and explainability—properties emphasised as essential in assistive contexts. Finally, the entire system is designed to run on modest hardware using local LLMs, avoiding cloud dependencies and supporting deployment in domestic environments where privacy, robustness, and transparency are paramount. By integrating lightweight implementation with structured memory management, SMILE occupies a distinct niche within the landscape of social robotics: a deployable, user-centred system capable of maintaining personalised relationships across sessions without the overhead of full cognitive architectures.

3 System Overview

SMILE (Social Memory-Integrated Language Environment) is designed as a hybrid cognitive architecture that combines multimodal perception, explicit memory structures, and a controlled language generation module. The system has been developed with the specific goal of supporting natural, personalised and long-term interactions with users, particularly older adults and individuals with cognitive vulnerabilities. The core principle guiding the design is that a robot intended for companionship must behave predictably and safely, while still maintaining an engaging conversational presence. In this section, we describe the full system pipeline, from audiovisual perception to dialogue management and memory integration. Rather than relying on the language model to implicitly “reason” about the state of the interaction, SMILE externalises memory, context handling and interaction flow, delegating to the LLM only the surface-level generation of linguistic responses. This separation drastically improves robustness, prevents hallucinations, and ensures that the robot behaves in a controlled and socially appropriate way. As discussed in Section 2, LLMs are powerful but fundamentally unstable: they do not retain internal state across turns, they often hallucinate memories and facts, and they are sensitive to noise from automatic speech recognition. In assistive robotics, where the perception of trust and coherence is essential, these behaviours cannot be tolerated. SMILE therefore adopts a hybrid architecture in which the robot—not the LLM—is responsible for memory, context control, and the overall interaction flow. The LLM is used exclusively as a *linguistic realiser*: it transforms a well-structured prompt into a natural and coherent utterance. All high-level reasoning, memory retrieval and dialogue management are determined externally through code. This design guarantees a stable dialogue experience and provides explicit control over when and how the robot should use information about the user.

3.1 High-Level Architecture

Figure 2 illustrates the key components of the system and their interactions. The robot integrates visual perception, speech processing, a deterministic dialogue manager, and multiple memory stores. The vision pipeline continuously monitors the environment, detecting faces and identifying users through FaceNet embeddings. When a user is recognised, the system loads the corresponding profile from persistent memory, allowing the robot to personalise the interaction from the first utterance. Speech is processed through a voice activity detector (VAD) and a speech recogniser. The VAD regulates turn-taking by identifying pauses in the audio stream, and the transcription is cleaned to remove noise or artefacts before entering the dialogue manager. This preprocessing step is crucial for reducing the impact of ASR errors on downstream components. The heart of the system is the Dialogue Manager (discussed in detail in Section 5), implemented as a finite-state machine (FSM). It orchestrates the development of the interaction and applies a small set of rules that regulate greetings, topic continuity, turn transitions, and conversational closure. Unlike LLM-driven state tracking, this FSM is deterministic and transparent, allowing precise

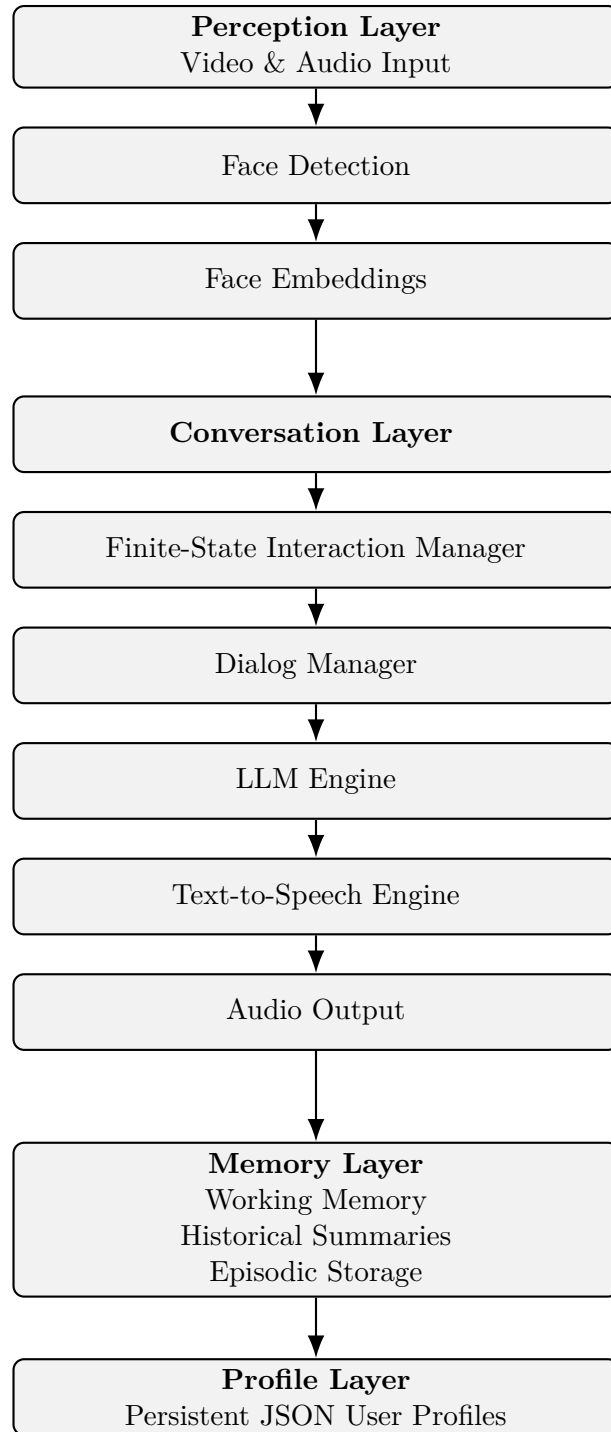


Figure 2: SMILE overall architecture. The entire pipeline is shown vertically from perception to persistent storage.

control over the robot’s behaviour in each phase of the interaction. The system maintains three complementary forms of memory, mirroring the taxonomy described in Section 2. A short-term *working memory* stores the most recent dialogue turns and is used to maintain local coherence. A *semantic memory* stores long-term stable information about the user, such as age range, interests, and personality traits. Finally, an *episodic memory* stores discrete experiences the user has narrated in past interactions, enabling the robot to recall and reuse personal stories only when explicitly relevant. The final stage of the pipeline is the prompt construction module. It consolidates the current conversation state, the user profile, selected episodes, recent dialogue history, and the user’s latest message into a structured textual prompt. This prompt is passed to a LLaMA-based LLM running locally via Ollama, which generates a concise, warm and contextually appropriate reply. Importantly, the LLM is not allowed to improvise or infer new memories: it can only operate on the information explicitly provided by the system through the structured prompt.

3.2 Multimodal Perception Pipeline

At the perceptual level, SMILE integrates vision and audio to identify users and capture their input. The vision module continuously samples frames from the camera and applies the MTCNN face detector [Zhang et al., 2016a]. Detected faces are encoded using FaceNet [Schroff et al., 2015] and compared to stored embeddings using cosine similarity. If the system identifies a known user, their profile is immediately loaded; otherwise, the robot initiates a name acquisition sequence and creates a new profile on the fly. This visual identification mechanism ensures that users do not need to explicitly re-identify themselves in subsequent sessions, supporting the naturalness of the interaction. On the audio side, the robot uses a tuned VAD to regulate turn-taking. Rather than relying solely on fixed timing thresholds, the system monitors RMS energy levels and silence durations to determine whether the user has finished speaking. Speech is transcribed using the Vosk offline ASR engine and normalised to a clean text form that serves as the basis for memory updates and LLM prompts. The use of offline ASR ensures privacy and reduces latency, both critical factors for deployment in home environments with vulnerable users.

3.3 Core System Components

SMILE’s architecture comprises four tightly integrated functional modules that work in concert to enable memory-augmented social interaction:

Dialogue Manager — A finite-state machine (Section 5) that explicitly controls conversational flow through three states (GREETING, FREE_TALK, FAREWELL), ensuring predictable state transitions and preventing common LLM failure modes such as greeting loops or unintended conversation prolongation.

Memory Subsystem — A three-layer architecture (Section 4) maintaining working memory (recent turns), semantic memory (stable user attributes), and episodic memory (discrete past experiences). Memory access is deterministic and mediated entirely by the dialogue manager rather than the LLM.

Prompt Construction Module — Integrates current dialogue state, selected memory fragments, behavioral constraints, and user input into a structured prompt that positions the LLM strictly as a linguistic realizer. All high-level reasoning occurs in code, not in the model.

Consolidation Pipeline — A post-session process (Section 5) that uses specialized LLM prompts to extract semantic attributes and episodic events from working memory, updating the persistent user profile with explicit grounding constraints to prevent hallucination.

The detailed design principles, access patterns, and consolidation mechanisms for each component are described in the following sections, with particular emphasis on how deterministic control and structured memory enable safe, transparent interaction suitable for assistive robotics contexts.

4 Memory Architecture

A fundamental challenge in social robotics is the ability to maintain coherent, long-term, personalised interactions with users. Unlike short-lived dialogues with conversational agents, a robot intended for companionship must preserve a sense of continuity across days, weeks and even months. This requirement demands a memory architecture that goes beyond simple context windows or database lookups. It must not only store information but *use* it in socially appropriate ways. SMILE approaches this challenge by implementing an explicit, structured memory system inspired by the cognitive memory taxonomy discussed in Section 2. As described, effective robot memory systems must be active, multi-modal, inherently episodic, associative, and introspectable. However, SMILE adapts these principles to the specific constraints of LLM-based dialogue systems and assistive robotics contexts, where simplicity, transparency, and predictability take precedence over computational sophistication. This section describes the conceptual framework and implementation of the three memory systems that jointly support SMILE’s behaviour: working memory, semantic memory and episodic memory. We discuss how these memory systems interact, when they are accessible to the dialogue manager, and how the system ensures safe, grounded and non-hallucinatory use of stored information.

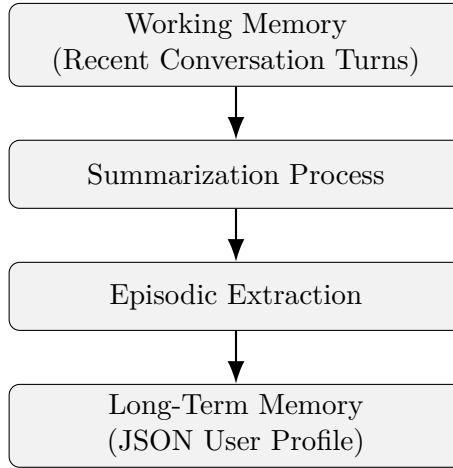


Figure 3: Flow between short-term and long-term memory in SMILE. Working memory is consolidated through LLM-based summarization, extracting both stable semantic attributes and discrete episodic events that are stored in the persistent user profile.

4.1 Design Principles

The design of SMILE’s memory subsystem follows three central principles derived from cognitive robotics and concerns raised in recent LLM memory literature (as discussed in Section 2):

1. **Separation of memory types.** Human memory is not monolithic: as illustrated in Figure 1, it is commonly modelled as a hybrid system involving working memory, long-term semantic knowledge, and episodic recollection of personal experiences. SMILE adopts this distinction explicitly, implementing separate structures for each memory type with distinct access patterns and persistence characteristics.
2. **Controlled access.** In social assistive robotics, uncontrolled recall can be intrusive, inappropriate or even distressing for vulnerable users. Therefore, memory is never queried directly by the LLM. Instead, the dialogue manager provides only the relevant retrieved elements, and only under specific conditions determined by explicit rules.

3. **Verifiable grounding.** Memory entries must be grounded in user statements. SMILE avoids the hallucination risks of end-to-end memory-augmented LLMs by ensuring that all stored information originates from the user’s own words, summarised and formalised by a post-conversation analysis module.

These principles ensure that memory improves conversational quality without compromising safety or reliability—essential requirements for long-term interaction with older adults or individuals with cognitive impairments.

4.2 Memory Components and Their Interactions

Figure 3 illustrates the flow of information between SMILE’s memory subsystems. Each component serves a distinct function while contributing to the system’s overall ability to maintain coherent, personalized interactions across multiple sessions.

4.2.1 Working Memory

Working memory constitutes the short-term context buffer of the system. SMILE maintains the last seven conversational turns, stored as a structured list of (**user**, **robot**) message pairs. This capacity reflects the constraints of human working memory discussed in Section 2 [Miller, 1956, Cowan, 2001]. The buffer supports local coherence (pronoun resolution, clarifications), context-dependent follow-up questions, and continuity in multi-turn exchanges. Crucially, this buffer is ephemeral: it resets at the end of each interaction. This design mirrors the role of working memory in human cognition, which supports ongoing tasks but does not retain information across episodes unless it is consolidated [Baddeley, 2012a]. In SMILE, consolidation occurs only when the post-session summariser identifies stable user attributes or salient episodic events worthy of long-term storage. Working Memory also protects the system from the instability of end-to-end LLM context windows. By strictly controlling what the LLM sees, the robot avoids the risk of the model reintroducing irrelevant topics or spurious details that may have appeared many turns earlier—a common failure mode in unconstrained LLM dialogue systems.

4.2.2 Semantic Memory

Semantic Memory stores long-term, stable knowledge about the user, including estimated age range, occupation or area of study, stable interests, personality descriptors, and long-term goals. These elements are stored as a structured JSON profile and updated only when explicit evidence emerges from the conversation. The summarisation module that maintains this memory operates conservatively: it integrates new attributes only when they are grounded in explicit user statements, rejecting inferences not directly supported by conversational evidence. Unlike working memory, semantic memory is not continuously shown to the LLM. The dialogue manager selects only the relevant stable attributes for prompt construction based on the current conversational topic. For instance, if the user discusses their research project, the robot may include information about their occupation or interests, but not irrelevant attributes such as unrelated personality traits or goals. This selective exposure ensures that the robot remains context-aware without appearing intrusive or over-dependent on stored personal information.

4.2.3 Episodic Memory

Episodic memory constitutes one of the most distinctive elements of the SMILE architecture. The system extracts and stores personal events mentioned by the user during conversation, representing each episode through a structured record containing a title, a temporal reference (e.g., “yesterday”, “last weekend”), a concise summary, and a set of semantic tags for retrieval. These episodes accumulate across interactions and provide a longitudinal narrative that enables

the robot to recall past experiences with the user. SMILE employs an explicit, rule-based retrieval mechanism. Episodic memories are incorporated into the LLM prompt only when the user’s current utterance overlaps with an episode’s tags or with semantically related keywords. This design mirrors the cue-dependent nature of human episodic recall, which is typically triggered by contextual signals rather than being continuously active [Tulving, 2002]. Similar principles have also been advocated in recent work on memory mechanisms for LLM-based agents, where episodic retrieval is treated as an event-driven process rather than a persistent background operation [Pink et al., 2025]. By externalising and constraining memory access, SMILE avoids the risks associated with automatic retrieval in LLM systems, such as hallucinated callbacks or inadvertent disclosure of sensitive information. The deterministic retrieval pipeline also ensures that memory usage remains explainable and auditable—qualities that are essential for assistive technologies deployed with vulnerable populations.

4.2.4 Memory System Coordination

Although the three memory types are stored separately, their interaction is coordinated through the dialogue manager. When a new user utterance is received, the system follows a structured retrieval process:

1. working memory is updated with the new conversational turn;
2. semantic memory is queried for stable attributes relevant to the current topic, using keyword matching against the user’s utterance;
3. episodic memory is searched for relevant episodes using tag-matching and keyword overlap heuristics;
4. the prompt construction module integrates only the selected memory fragments into the final LLM prompt, along with explicit instructions about how to use this information.

This approach enhances transparency, reduces the risk of spurious associations, and ensures that memory usage can be explained and audited. The trade-off is reduced flexibility compared to learned retrieval, but for assistive robotics applications, predictability and safety take precedence over sophistication. The consolidation process shown in Figure 3 occurs at the end of each session. The system uses the LLM in a controlled way to summarize the working memory buffer, extracting stable semantic attributes and discrete episodic events. This summarization prompt is carefully designed to prevent hallucination: the LLM is explicitly instructed to extract only information directly stated by the user and to reject any inferences or assumptions. The resulting updates are then merged with the existing user profile, with conflicts resolved by prioritizing the most recent information.

5 Dialogue Management

The dialogue manager is the core component that determines how SMILE behaves moment by moment during an interaction. While the memory architecture described in Section 4 provides the long-term and short-term knowledge structures, it is the dialogue manager that decides *when* and *how* these memories should be used. The system is designed to behave as a socially appropriate conversational partner, particularly for use cases involving long-term interaction with older adults or individuals with cognitive impairments. For this reason, SMILE adopts an explicit, deterministic dialogue management strategy rather than leaving control to the language model. This design choice directly addresses the concerns raised by Irfan et al. [Irfan et al., 2025] regarding the inconsistency and unpredictability of purely LLM-driven dialogue systems. By externalizing dialogue control, SMILE ensures predictability, safety and interpretability—essential properties for assistive robotics applications.

5.1 Finite State Machine Architecture

As depicted in Figure 4 the dialogue manager operates as a finite state machine (FSM) that governs the structure of each conversational session. At runtime, the system continuously processes three types of input:

1. the recognised user utterance (via ASR),
2. the current conversation state,
3. the relevant portions of memory retrieved by the memory subsystem.

Based on these inputs, it constructs a tailored prompt, queries the language model and generates an audio response via the TTS module. The explicit FSM structure provides the backbone for conversational flow and prevents the LLM from producing inappropriate or repetitive behaviours.

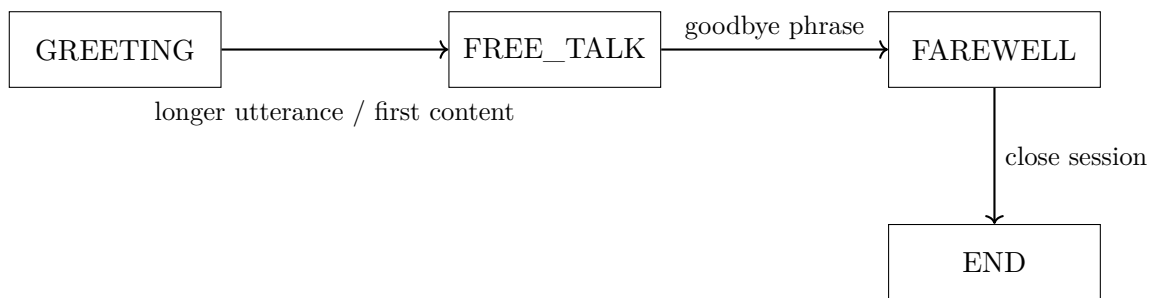


Figure 4: Finite State Machine governing SMILE’s conversational flow.

5.1.1 State Definitions and Transitions

The conversation is governed by three main states, as illustrated in Figure 2:

- **GREETING** — Triggered each time the system recognises a known user (via face recognition) or registers a new one. The robot is allowed to use the user’s name and perform a socially appropriate opening. Memory access is minimal in this state to avoid overwhelming the user with stored information during initial contact.
- **FREE_TALK** — The main open-ended conversational state. Memory access is fully enabled, topic-following is active, and the robot maintains context coherence while adhering to strict behavioural constraints. This state reflects the principles of working memory management discussed in Section 4.
- **FAREWELL** — Entered when the user expresses an explicit intention to end the interaction. The robot produces a brief, warm goodbye without prolonging the conversation, respecting the user’s desire to terminate the session.

The transitions between these states are explicit and deterministic:

- **GREETING → FREE_TALK**: Triggered as soon as the user produces more than a minimal salutation or begins expressing substantive content.
- **FREE_TALK → FAREWELL**: Triggered when the user’s utterance contains lexical indicators such as “I have to go”, “see you later”, “goodbye”, and similar closing expressions detected through keyword matching.

- **END:** Once the FAREWELL response is generated, the session terminates and consolidation (memory update) occurs as described in Section 4.

This explicit state control prevents the LLM from misinterpreting a closing sentence as a new topic or from defaulting into unwanted greetings during long sessions.

5.2 Behavioural Constraints and Prompt Engineering

Within each state, the dialogue manager enforces a set of strict behavioural constraints through careful prompt construction. The prompt operates as a behavioural “contract” for the LLM, ensuring that responses remain consistent, contextual and aligned with interaction norms appropriate for vulnerable populations. Key rules embedded in the prompt include:

- No repeated greetings outside of the GREETING state
- Replies limited to short, concise 2–3 sentence responses to avoid cognitive overload
- No unsolicited topic changes that might confuse or disorient the user
- No emotional mind-reading or assumptions not grounded in explicit user input
- Confirmation questions in case of unclear or possibly misrecognised words
- Selective and contextual use of memory (working, semantic, episodic) as determined by the dialogue manager

By providing these constraints explicitly to the LLM through structured prompting, the system avoids common pitfalls of conversational generative systems, such as overly verbose replies, hallucinated emotional interpretations, or unwanted shifts in topic. This approach is consistent with best practices for LLM-based assistive technologies Wei et al. [2023], where control and predictability are paramount.

5.3 Memory Integration and Retrieval Control

Memory access is entirely mediated by the dialogue manager, following the principles of controlled access outlined in Section 4. The LLM never retrieves memory independently; instead, the manager constructs a prompt that includes only:

1. Relevant elements of the user’s semantic memory (e.g., occupation, interests) filtered based on topic relevance to the current utterance
2. The last seven turns of working memory, maintaining the capacity constraints discussed in Section 2 [Miller, 1956, Cowan, 2001]
3. Only those episodic memories whose tags or summaries overlap with keywords in the current user message

As previously mentioned this design ensures contextual coherence without over-sharing personal information, safety for vulnerable populations (no unwanted or distressing reminders), grounding in user-verified information only and explainability of why a particular memory was recalled. The deterministic control exerted by the dialogue manager strongly differentiates SMILE from end-to-end memory-augmented LLM architectures discussed in Section 2.

5.4 Prompt Construction and Response Generation

For each user utterance, the dialogue manager executes a structured pipeline:

1. Identify the current state (GREETING, FREE_TALK, FAREWELL)
2. Apply state-specific behavioural rules
3. Select relevant memory fragments using deterministic retrieval
4. Construct the full prompt incorporating:
 - State description and transition context
 - Behavioural constraints and response guidelines
 - Filtered semantic memory attributes
 - Contextually matched episodic memories (if any)
 - Working memory trace (last 7 turns)
 - Current user utterance
5. Query the local LLM (via Ollama)
6. Post-process the generated response:
 - Remove any unwanted greeting phrases
 - Enforce conciseness constraints
 - Filter repetitive content
7. Dispatch the final reply to the TTS module

This pipeline is explicitly implemented in code and does not depend on the internal behaviour of the LLM, ensuring reproducibility and robustness across different conversational contexts. The separation of concerns between dialogue management (handled by deterministic code) and linguistic realization (handled by the LLM) reflects the hybrid architecture principles advocated by cognitive robotics research [Peller-Konrad et al., 2023]. The complete prompt template with inline commentary is provided in Appendix C, demonstrating how dialogue state, memory fragments, and behavioral constraints are encoded in natural language instructions to the LLM.

5.5 Error Handling and Robustness

Real-world audio interaction introduces significant uncertainty, particularly due to speech recognition errors. The dialogue manager incorporates several strategies for ambiguity resolution and error recovery:

- Unclear utterances: request explicit confirmation rather than guessing user intent;
- Potential misrecognition: if a suspected ASR error introduces a false semantic shift, the system ignores it unless the user repeats the same content;
- Hallucination prevention: refuse to assume facts not present in memory or explicitly stated by the user;
- Topic coherence: maintain conversation thread even when ASR introduces noise, using working memory context to disambiguate.

These error-handling behaviours are crucial for settings involving older adults or cognitively impaired users, where misinterpretations may cause confusion, frustration or distress. The system’s conservative approach prioritizes reliability over conversational fluency when faced with uncertainty.

5.6 Post-Conversation Consolidation

Once the conversation ends (upon entering the FAREWELL state), SMILE enters a dedicated consolidation phase. The working memory buffer is processed by a specialized summarization prompt that:

- Extracts new stable attributes for semantic memory (e.g., newly mentioned interests, life events);
- Identifies discrete episodic events suitable for long-term storage;
- Updates the global conversation summary;
- Integrates new tags and keywords into the user’s profile;
- Explicitly rejects hallucinated or inferred attributes not directly stated by the user.

This offline consolidation process parallels the memory consolidation observed in human cognition, where episodic experiences are transformed into stable semantic knowledge through a reflective process that occurs after the original experience [Tulving, 2002]. In cognitive robotics, similar architectures advocate for separating online interaction from reflective memory encoding [Peller-Konrad et al., 2023], a principle that SMILE adopts to ensure memory quality and prevent the accumulation of erroneous information. The consolidation prompt is carefully designed to be conservative: it includes explicit instructions to extract only information directly stated by the user and to mark any uncertainty. This grounding requirement ensures that the user profile evolves safely over time, reflecting actual user disclosure rather than system inference.

6 System Implementation

This section describes the technical implementation of SMILE, demonstrating how face recognition, speech processing, dialogue management, and memory consolidation integrate into a coherent real-time pipeline. The system is implemented entirely in Python and designed to run on commodity hardware with GPU acceleration, enabling real-time multimodal interaction suitable for deployment in assistive robotics contexts.

6.1 Modular Software Architecture

The implementation follows a layered, modular design where each component addresses a distinct perceptual or cognitive function: the entire loop is depicted in Figure 5 and resides in `recognize_live.py`, which coordinates five functional layers (see Appendix A.1):

- Vision — face detection, tracking, and embedding-based identification;
- Audio — continuous speech recognition (ASR) and text-to-speech (TTS);
- Dialogue — finite-state machine governing conversational flow;
- Memory — working, semantic, and episodic memory subsystems;
- Language — local LLM inference via Ollama (llama3).

This separation of concerns ensures modularity, testability, and transparent control over the robot’s behavior—critical properties for systems deployed with vulnerable populations.

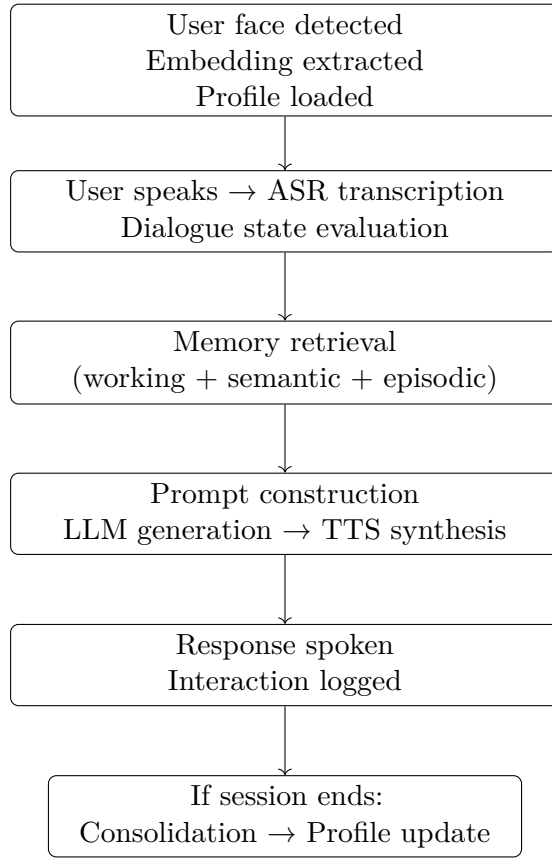


Figure 5: Complete interaction cycle in SMILE, from perception through response generation to memory consolidation.

6.2 Multimodal Perception Pipeline

6.2.1 Face Recognition and User Identification

The vision module continuously processes webcam frames using MTCNN for face detection Zhang et al. [2016b] and extracts 128-dimensional embeddings via a pre-trained FaceNet model Schroff et al. [2015]. User identification proceeds through cosine similarity comparison against stored embeddings; matches above a threshold of 0.6 trigger profile retrieval, while unrecognized faces initiate a registration dialogue to acquire the user’s name. Face processing runs asynchronously in a dedicated thread to prevent blocking the main interaction loop. This design ensures the dialogue remains responsive even during computationally intensive recognition operations—a requirement for natural real-time interaction Bohus and Rudnicky [2009].

6.2.2 Speech Processing

Audio capture employs a continuous microphone stream with adaptive silence detection to regulate turn-taking. The system monitors RMS energy levels and applies a dynamic threshold to determine speech endpoints, addressing the challenge of natural pause detection in spontaneous conversation Raux and Eskenazi [2008]. Once silence is detected, the audio buffer is transcribed using Vosk offline ASR¹, avoiding cloud dependencies that raise privacy concerns in assistive applications. The transcribed text undergoes minimal normalization (lowercasing, punctuation removal) before entering the dialogue manager. Speech synthesis uses `pyttsx3` with asynchronous execution: TTS generation occurs in a background thread, allowing the system to resume listening immediately after initiating audio output. This overlapping execution eliminates perceptible delays between user input and robot response.

6.3 Dialogue Management and LLM Integration

6.3.1 Finite-State Conversational Control

The dialogue manager (`dialog_manager.py`) implements the three-state FSM described in Section 5: `GREETING`, `FREE_TALK`, and `FAREWELL`. State transitions are triggered by explicit lexical patterns (e.g., goodbye phrases for `FREE_TALK` → `FAREWELL`) rather than LLM inference, ensuring deterministic and explainable behavior. At each turn, the manager constructs a structured prompt incorporating:

1. Current dialogue state and associated behavioral constraints;
2. Filtered semantic memory attributes (age, occupation, interests);
3. Contextually matched episodic memories (via tag-based retrieval);
4. Recent working memory trace (last 7 turns);
5. Temporal grounding (current date and time);
6. User’s utterance.

Appendix A.2 provides the complete implementation of the prompt construction logic, including how semantic, episodic, and working memory are selectively integrated based on dialogue state. The prompt explicitly constrains LLM behavior through meta-instructions that prohibit repeated greetings, unsolicited topic changes, emotional assumptions, and verbose responses. This approach reflects best practices in prompt engineering for controlled generation Irfan et al. [2025], adapted to the specific requirements of assistive social robotics.

¹The system uses the Italian Vosk model `vosk-model-it-0.22` for privacy-preserving local inference.

6.3.2 Local LLM Inference

SMILE uses Ollama to host `llama3` locally, avoiding the latency, cost, and privacy concerns of cloud-based APIs. The model receives the constructed prompt and generates a response constrained to 2–3 sentences. Post-processing filters remove any residual greeting phrases or repetitive content before TTS synthesis.

6.4 Memory Persistence and Consolidation

User profiles are stored as structured JSON files in `data/profiles/`, each containing semantic attributes (name, age range, interests, personality descriptors) and an episodic memory array (titled events with timestamps, summaries, and retrieval tags). Working memory exists only in RAM and resets after each session, mirroring the ephemeral nature of human working memory Baddeley [2012b]. At session termination, the consolidation module (`memory_manager.py`) invokes a specialized summarization prompt that processes the working memory buffer to extract new or refined semantic attributes explicitly stated by the user, discrete episodic events suitable for long-term storage and finally Updated conversational summary for the session. The consolidation prompt includes strict grounding instructions: the LLM must extract only information directly supported by user utterances and mark any uncertainty. This conservative approach prevents hallucinated attributes from contaminating the profile.

6.5 Concurrency and Performance Optimization

The implementation employs thread-based concurrency to maintain real-time responsiveness despite computationally intensive operations. Face recognition, TTS synthesis, and LLM inference execute in separate threads or use asynchronous calls, preventing any single component from blocking the main interaction loop. GPU acceleration (CUDA) is used for face embedding extraction and, where available, ASR inference. Warm-up routines at system startup preload models and execute dummy inferences to eliminate first-call latency, ensuring smooth interaction from the first user encounter.

6.6 Transparency and Reproducibility

All interactions are logged to disk in human-readable JSON format, capturing user utterances, robot responses, memory updates, and state transitions. This comprehensive logging enables:

- Post-hoc analysis of conversational dynamics;
- Longitudinal studies tracking profile evolution over weeks or months;
- Clinical validation for applications in cognitive assessment or therapy;
- Complete reproducibility of system behavior for debugging or research.

The architecture was explicitly designed for *introspectability* Peller-Konrad et al. [2023]: every memory change, retrieval decision, and LLM interaction can be reconstructed from the logs, supporting both technical debugging and scientific scrutiny of long-term human-robot interaction patterns.

7 Evaluation

Evaluating memory-augmented conversational systems for social robotics presents unique methodological challenges. Unlike task-oriented dialogue systems where success can be measured through completion rates, or open-domain chatbots evaluated primarily on linguistic quality, companion robots require assessment along multiple dimensions that span technical performance, user experience, and memory coherence. This section reviews established evaluation methodologies in related domains and proposes a practical evaluation protocol tailored to SMILE’s hybrid architecture. An example of SMILE’s output—including an extracted user profile and annotated conversation transcript—is provided in Appendix B, illustrating concrete memory retrieval patterns and consolidation results.

7.1 Evaluation Practices in Conversational AI and Social Robotics

7.1.1 Memory and Personalization Assessment

The evaluation of memory systems in conversational agents has received increasing attention in recent years Zhang et al. [2025]. Effective evaluation must distinguish between different temporal scales of memory: turn-level memory (maintaining coherence within immediate responses), conversation-level memory (retaining information throughout a session), and long-term persistence across multiple encounters. A critical metric for conversational memory is *knowledge retention*, which assesses how well a system avoids asking repetitive questions about information users have already provided. This directly addresses a common source of frustration in long-term human-robot interaction, where users expect the robot to remember previous conversations and personalize interactions without treating each session as an isolated event Irfan et al. [2025].

7.1.2 Dialogue Quality Metrics

Traditional automatic metrics such as BLEU Papineni et al. [2002] and ROUGE Lin [2004] have been shown to correlate weakly with human judgments of dialogue quality, as they measure only superficial linguistic properties rather than conversational dynamics. Modern evaluation paradigms instead focus on dimensions including:

- Coherence — Contextual appropriateness and logical consistency;
- Engagement — Capacity to sustain natural conversation flow;
- Informativeness — Providing relevant and useful content;
- Consistency — Avoiding contradictions, particularly with stored memory;
- Factual correctness — Grounding responses in verified information.

Recent approaches leverage LLMs themselves as evaluators, using structured prompts to assess dialogue quality along these dimensions. However, such automated evaluation methods require careful validation against human judgment, particularly for systems designed for vulnerable populations.

7.1.3 Usability and Acceptability in HRI

For social robots deployed with users, standardized instruments such as the System Usability Scale (SUS) Brooke [1996] have proven effective for tracking user perception across deployment phases. The SUS provides a reliable measure of perceived usability through ten questions rated on a 5-point Likert scale, yielding scores from 0 to 100. Scores above 70 are generally considered indicative of good usability, while scores below 50 suggest significant usability problems. Beyond

usability, *acceptability*—the degree to which users are willing to engage with a system over time—is crucial for social robots. While longitudinal deployment studies remain the gold standard for assessing sustained acceptance, initial acceptability can be evaluated through post-interaction questionnaires and semi-structured interviews.

7.2 Evaluation Challenges Specific to Memory-Augmented Systems

SMILE’s architecture introduces specific evaluation requirements related to memory accuracy, retrieval appropriateness, and profile evolution:

- **Memory extraction accuracy:** Do semantic attributes and episodic events extracted during consolidation accurately reflect user statements?
- **Retrieval appropriateness:** Are episodic memories retrieved at contextually relevant moments without intrusive over-sharing?
- **Profile consistency:** Do user profiles remain coherent over multiple sessions, or do contradictions accumulate?
- **Recognition reliability:** Does face-based identification work consistently across varying conditions (lighting, camera angle, appearance changes)?

7.3 Proposed Evaluation Protocol

Given the practical constraints of a research prototype and the need to demonstrate core functionality before longitudinal deployment with target populations, we propose a two-phase evaluation approach that balances methodological rigor with feasibility.

7.3.1 Phase 1: Technical Validation and Usability Assessment

Objectives: Evaluate core system functionality (face recognition, memory extraction, dialogue coherence) and establish baseline usability indicators through controlled interaction sessions.

Participants: 15–20 adults recruited from the university community (students, staff, faculty) with diverse demographics (age 20–60+, varied technology familiarity). Although this population differs from the eventual target demographic of older adults, it enables rapid iteration on technical and interface-level aspects before conducting specialised studies.

Protocol: Each participant engages in three sessions over a two-week period:

1. **Session 1 (Initial encounter, 15 min):** System registers the user using face recognition, acquires their name through dialogue, and collects initial personal information. *Measured outcomes:* face registration success, name acquisition accuracy, initial profile creation.
2. **Session 2 (One week later, 10–15 min):** System should recognise the returning user and personalise the greeting. Conversation progresses with new topics while referencing previously discussed content. *Measured outcomes:* recognition accuracy, greeting personalisation, semantic recall, profile updates.
3. **Session 3 (Two weeks after Session 1, 10–15 min):** System engages in conversation including references to events from prior sessions. *Measured outcomes:* episodic retrieval appropriateness, long-term profile consistency, multi-session coherence.

After the final session, participants complete:

- the System Usability Scale (SUS);

- a post-interaction questionnaire (5-point Likert scales) assessing: perceived conversational naturalness, appropriateness of memory use (“felt the robot knew me”), comfort with stored information, and likelihood of continued interaction;
- a brief semi-structured interview exploring moments where memory use felt helpful or intrusive, technical issues, and suggestions for improvement.

Technical Metrics: Interaction logs (Section 6) enable post-hoc evaluation of:

- Face recognition performance: overall identification accuracy, false acceptance and false rejection events across sessions;
- Memory extraction accuracy: two independent raters annotate conversation transcripts and extracted profiles. Inter-rater reliability is computed using Cohen’s κ , a standard statistic for measuring agreement beyond chance. Agreement between human annotations and system outputs indicates the reliability of SMILE’s consolidation process;
- Knowledge retention: proportion of conversational turns in which SMILE correctly references previously mentioned information without re-asking;
- Retrieval appropriateness: for each episodic recall event, human coders evaluate whether the retrieved memory is relevant to the ongoing conversational context;
- Profile consistency: analysis of contradictions or inaccuracies accumulated across sessions.

Baseline Comparisons: To isolate the contribution of memory, SMILE can be compared against two baselines:

1. a **stateless baseline** where each session is treated as a first encounter, controlling for LLM behaviour independently of memory;
2. a **simple logging baseline** that stores raw conversation history but performs no summarisation or structured extraction, allowing the LLM to access prior turns without organisation.

A within-subjects design, with counterbalanced order of SMILE and baseline conditions, enables direct comparison of perceived personalisation, usability, and memory-related interaction quality.

7.3.2 Phase 2: Extended Interaction and Profile Evolution

Objectives: Assess memory system behavior over longer timescales and validate profile evolution mechanisms.

Protocol: Recruit 5–8 volunteers for extended participation (6–8 sessions over 4–6 weeks, 2 sessions per week). This smaller cohort enables deeper qualitative analysis of how profiles develop and whether memory accumulation causes issues (contradictions, over-specificity, retrieval failures).

Analysis:

- Track profile growth: semantic attribute count, episodic memory count, summary length
- Analyze memory retrieval patterns: frequency of episodic recalls, user reactions (coded from transcripts)
- Conduct exit interviews exploring overall experience, trust in the system, comfort with accumulated data, and suggestions for memory management (e.g., should users be able to view/edit profiles?)

This phase provides preliminary evidence of longitudinal viability without requiring full-scale in-situ deployment, serving as proof-of-concept for future studies with target populations (older adults, individuals with MCI) that would necessitate ethical approval and specialized recruitment.

7.4 Limitations and Future Directions

The proposed evaluation has several limitations that constrain generalizability:

Population: validation with university community members differs from deployment with elderly users or individuals with cognitive impairments. Cognitive load, interaction preferences, and acceptance criteria may differ substantially. However, this initial validation is essential to identify and resolve technical issues before engaging vulnerable populations.

Duration: two-week (Phase 1) and 6-week (Phase 2) timescales are insufficient to assess true long-term engagement and profile evolution over months or years. Longitudinal studies spanning 3–6 months with target populations remain a critical future direction.

Ecological validity: controlled laboratory sessions differ from naturalistic home deployment. Real-world factors including ambient noise, variable lighting, spontaneous interaction patterns, and integration into daily routines require in-situ studies that fall outside the scope of initial validation.

Absence of Baselines: while the evaluation proposes within-subjects comparison against stateless and simple-logging baselines, resource constraints may limit implementation of full comparison systems. Without direct head-to-head evaluation against existing memory-augmented dialogue systems or commercial companion robots, SMILE’s relative performance remains uncertain. Future work should prioritize comparative studies using standardized interaction scenarios and shared evaluation metrics, enabling the research community to systematically assess trade-offs between different memory architectures (deterministic vs. RAG-based, prompt-controlled vs. algorithmically-gated, etc.).

Despite these limitations, the proposed protocol provides essential evidence of SMILE’s core capabilities and user acceptability, establishing a foundation for subsequent deployment with assistive care populations. Future work should extend evaluation to:

- Longitudinal field studies with older adults in residential or assisted living settings
- Clinical validation for specific populations (mild cognitive impairment, post-hospitalization recovery)
- Comparative studies against commercial companion robots
- Multi-user scenarios requiring shared versus individual memory management
- Integration of emotion recognition for affective memory tagging

In summary, evaluating SMILE requires balancing the need for rigorous technical validation with the practical constraints of prototype development. The proposed two-phase approach provides systematic assessment of memory accuracy, dialogue coherence, and user acceptance while remaining feasible within typical research timelines and resources. This initial validation establishes the empirical groundwork necessary to justify and design future longitudinal studies with the system’s intended user populations.

8 Discussion and Limitations

The development of SMILE demonstrates both the potential and the inherent constraints of hybrid memory architectures for embodied conversational agents. The system successfully integrates face recognition, structured memory persistence, and LLM-based dialogue generation into a functioning prototype suitable for assistive robotics contexts. However, the design and implementation process revealed several conceptual and technical limitations that warrant careful examination. Many of these challenges reflect broader open problems in memory-augmented social robotics and suggest directions for future research.

8.1 Architectural Trade-offs: Determinism versus Flexibility

SMILE’s architecture embodies a deliberate commitment to deterministic, transparent mechanisms for dialogue control and memory access. This approach prioritizes safety, explainability, and predictable behaviour—properties that are particularly important in assistive contexts. However, these advantages come at the cost of reduced flexibility compared to systems that grant the LLM greater autonomy. A central trade-off emerges from SMILE’s reliance on structured prompts and state-based logic to constrain LLM behaviour. Although prompt engineering can encode high-level behavioural rules, such instructions function as *soft constraints*. They can be overridden by conversational drift, conflicting user requests, or noise introduced by automatic speech recognition. As noted in work on instruction-following models, natural-language prompts lack mechanisms for strict priority enforcement or guaranteed compliance under all conditions [Ouyang et al., 2022]. SMILE’s finite-state dialogue manager mitigates some risks—such as inappropriate topic shifts or repeated greetings—but cannot fully regulate fine-grained behaviours such as preventing episodic recall in borderline cases or enforcing rigid stylistic choices. This tension between *controllability* and *conversational naturalness* is pervasive in LLM-driven systems. More algorithmic forms of control (e.g., constrained decoding, schema-based response formats) could strengthen determinism but risk reducing fluidity and user engagement. SMILE thus adopts a conservative middle ground that ensures safety and transparency, while accepting that some behaviours will remain probabilistic.

8.2 Memory Representation Challenges

SMILE employs a classical division between semantic and episodic memory: semantic memory captures stable personal attributes, preferences, and biographical details, whereas episodic memory stores discrete events mentioned by the user with explicit temporal grounding. While architecturally clean, this division introduces practical challenges during real-time interaction and post-session consolidation.

8.2.1 The Semantic–Episodic Boundary

User utterances often blend stable preferences with contextually situated experiences. Statements such as “*I’ve been really into Cuban music lately*” can ambiguously signal a long-term interest, a temporary curiosity, or a recent episodic event. SMILE currently resolves such ambiguity through LLM-based consolidation rules: stable-sounding preferences are stored semantically, while time-bound descriptions become episodic records. However, these heuristics occasionally lead to inconsistent classifications, especially when users revisit topics across sessions. This difficulty is consistent with cognitive findings that episodic memories tend to “semanticize” over time as contextual information decays and only generalized knowledge persists [Baddeley, 2012b]. Artificial systems that aim to mirror human-like memory organization must therefore account for evolving stability, identifying when repeated mentions justify transitioning an attribute from episodic to semantic memory. Doing so requires either explicit rules or longitudinal data-driven models, which lie beyond SMILE’s current scope but are planned for future evaluation.

8.2.2 Semantic Memory Stability

Even when semantic updates are unambiguous, challenges arise due to the LLM’s propensity to generalize from sparse evidence. Single conversational mentions may lead to premature or unwarranted profile updates (e.g., inferring core interests from casual remarks). Conversely, SMILE’s conservative design avoids inferring implicit attributes (such as lifestyle or personality traits) unless explicitly stated, occasionally missing meaningful personalization cues. These issues exemplify a fundamental tension: aggressive inference increases personalization at the

risk of hallucination, while conservative policies preserve safety and privacy but reduce adaptive richness.

8.3 Retrieval Mechanisms: Cognitive Plausibility versus System Controllability

Memory retrieval is arguably the most consequential aspect of SMILE’s architecture. It determines *when* past information influences interaction and shapes the user’s perception of personalization, attentiveness, and coherence. For this reason, SMILE adopts a deterministic, rule-based retrieval strategy designed to ensure transparency and prevent unpredictable recall behaviours. Episodic memories are surfaced only when their tags overlap with lexically or semantically recognizable elements in the user’s current utterance. Semantic attributes are similarly filtered by explicit relevance heuristics. This approach guarantees that memory activation is explainable: developers and users can trace exactly why the system retrieved a particular episode or attribute. It also prevents hallucinated callbacks, accidental disclosure of sensitive information, and contextually inappropriate reminders—failure modes commonly observed in LLMs with uncontrolled retrieval strategies. While safe and predictable, SMILE’s retrieval strategy diverges substantially from models of human episodic recall. Decades of cognitive research show that episodic memory retrieval is *cue-dependent*, *contextual*, and *graded*. Classical accounts emphasize that episodic recall is not a verbatim replay but a reconstructive process shaped by current goals, context, and affective states [Tulving, 2002, Schacter, 2011]. Contextual activation models, such as the CMR model [Polyn et al., 2009] and the Retrieved Context framework [Howard and Kahana, 2002], explain retrieval as a competition among memory traces whose activation strengths vary continuously with semantic, temporal, and emotional similarity to the present moment. Neural models further highlight the role of competitive inhibition and activation dynamics in shaping retrieval outcomes [Norman et al., 2005]. By contrast, SMILE retrieves memories only when a surface-level match occurs. This limits its ability to:

- respond to *implicit cues* (e.g., emotional states, thematic continuity, or goals not expressed lexically);
- prioritize memories based on *recency* or *emotional salience*;
- exploit *analogical similarity* across experiences;
- exhibit spontaneous reminiscence triggered by shifts in context.

Users may thus perceive SMILE as attentive but somewhat rigid, missing opportunities for more subtle or naturalistic recall. More flexible approaches to retrieval have been explored in the LLM-agent and memory-augmented NLP literature:

- **Embedding-based retrieval** activates memories via vector similarity rather than keywords [Zhang et al., 2025], enabling associative recall.
- **Reinforcement-learned retrieval policies** select memories that optimize downstream conversational success [Lampinen et al., 2021].
- **Hybrid rule-based and learned ranking** schemes balance safety constraints with richer prioritization [Madotto et al., 2021].

However, these methods introduce new risks: opaque reasoning, vulnerability to ASR-induced semantic distortion, difficulty in controlling sensitive information, and reduced explainability—practical barriers for assistive settings involving vulnerable users. SMILE therefore adopts a conservative but principled retrieval design that privileges safety and auditability over cognitive

fidelity. This choice aligns with the system’s assistive focus but highlights a key avenue for future work: developing hybrid retrieval mechanisms that remain transparent and controllable while supporting more context-sensitive, human-like memory activation.

8.4 Memory Extraction and Consolidation Challenges

While retrieval concerns govern how memory is used during interaction, extraction and consolidation determine *what* information becomes part of long-term memory in the first place. SMILE’s post-session consolidation process relies on LLM-guided analysis of working memory transcripts to identify episodic events and update semantic profiles. Despite strong results in prototype testing, several limitations remain. The episodic summarization prompt can produce inconsistent granularity, from overly broad descriptors (“*talked about hobbies*”) to excessively fine-grained micro-episodes. Users frequently reference events with vague temporal language (“*recently*”, “*a while ago*”), which complicates temporal normalization and weakens time-sensitive retrieval strategies. ASR errors can propagate into summaries, producing distorted or ambiguous episodes. Finally, SMILE does not yet merge episodes that refer to the same event across sessions, preventing richer, multi-faceted episodic representations from forming. Potential improvements include multi-stage consolidation pipelines, confidence estimation for extracted episodes, and structured temporal representations (e.g., ISO date ranges). These enhancements require extended longitudinal data and are therefore left to future evaluation phases.

8.5 Working Memory Limitations and Context Management

SMILE maintains a working memory of the last seven conversational turns, inspired by well-established constraints on human working memory capacity [Baddeley, 2012b]. While this design supports local coherence and reduces hallucination risks, it limits long-distance thematic tracking and the system’s ability to sustain complex multi-session narratives. Techniques such as hierarchical summarization, topic-aware context management, or similarity-based retrieval of past turns may extend the effective context window, but each introduces complexity and must be evaluated empirically.

8.6 Multimodal Perception Robustness

8.6.1 Speech Recognition Dependencies

SMILE’s exclusive reliance on speech interaction creates a critical dependency on ASR accuracy. While Vosk provides robust offline transcription for Italian, several failure modes persist:

- **Homophones and near-homophones:** Words with similar pronunciation produce transcription ambiguities that propagate directly into dialogue context and memory consolidation.
- **Accents and speech patterns:** Although Vosk supports dialectical variation, users with strong regional accents or atypical speech patterns (common in elderly populations or individuals with neurological conditions) experience elevated error rates.
- **Environmental noise:** Ambient sound in home environments degrades transcription quality, particularly for soft-spoken users or when the robot is positioned at a distance.

SMILE’s prompt includes instructions to request clarification when input seems garbled, and the dialogue manager filters obviously malformed transcriptions. However, *semantically plausible but incorrect* transcriptions are difficult to detect automatically and can lead to cascading errors in memory and response generation. Confidence-based filtering (discarding low-confidence ASR outputs), active confirmation for critical information (names, dates, important events),

or multimodal input (allowing users to type corrections via a companion interface). However, adding interaction modalities increases system complexity and may reduce the naturalness that speech-only interaction provides. The evaluation protocol’s technical metrics (Section 7) include logging ASR confidence scores alongside transcriptions, enabling post-hoc correlation analysis between recognition quality and downstream system performance.

8.6.2 Face Recognition Constraints

Vision-based user identification works reliably under controlled conditions but faces challenges in real-world deployment:

- **Appearance changes:** Glasses, facial hair, significant weight changes, or aging over extended deployment periods can reduce embedding similarity below recognition thresholds.
- **Lighting sensitivity:** MTCNN face detection performance degrades in low-light or high-contrast environments common in home settings.
- **Multi-user scenarios:** SMILE currently handles one recognized user at a time. Family environments where multiple people may appear in frame require disambiguation logic not yet implemented.

While Phase 1 evaluation (Section 7) will quantify recognition accuracy under varying conditions, addressing these issues fundamentally requires either more robust embedding models (with associated computational costs) or fallback identification methods (e.g., voice recognition, explicit name confirmation).

8.7 Ethical and Deployment Considerations

8.7.1 Privacy and Data Transparency

SMILE stores detailed personal information (conversation histories, inferred attributes, episodic memories) in local JSON files. While this avoids cloud dependencies and associated privacy risks Irfan et al. [2025], several concerns remain:

- **Informed consent:** Users may not fully understand what information is being stored or how it will be used, particularly elderly individuals with limited technical literacy.
- **Data persistence:** SMILE currently provides no mechanisms for users to view, edit, or delete their profiles. While this simplifies implementation, it conflicts with GDPR principles and user agency expectations Irfan et al. [2025].
- **Implicit inference:** Even with conservative extraction heuristics, the system inevitably infers attributes (e.g., classifying someone as "elderly" based on stated retirement) that were not explicitly disclosed.

Future iterations must implement user-facing transparency tools: profile viewers, selective memory deletion, and explicit consent workflows for different information types. The Phase 1 evaluation’s post-interaction questionnaire (Section 7) includes items assessing comfort with stored information, providing initial user perspectives on these issues.

8.7.2 Behavioral Safety in Assistive Contexts

SMILE’s target population—older adults and individuals with cognitive impairments—requires particularly cautious system design. The current implementation enforces strict behavioral constraints (no emotional mind-reading, no unsolicited topic changes, concise responses), but edge cases remain:

- **Contradictory user statements:** When a user provides information conflicting with their stored profile (e.g., previously stated being widowed, later mentioning a spouse), SMILE updates the profile without explicitly addressing the contradiction. For users with memory impairments, this could cause confusion or distress.
- **Sensitive disclosures:** Users may spontaneously mention traumatic events, health crises, or family conflicts. SMILE’s general-purpose conversational prompts do not include specialized protocols for such disclosures, risking inappropriate responses.
- **Over-reliance and attachment:** Extended interaction may lead vulnerable users to develop emotional dependence on the robot, potentially reducing human social contact—an outcome counter to assistive care goals.

These concerns underscore why Irfan et al.’s work Irfan et al. [2025] emphasizes the need for extensive pilot testing and iterative refinement with target populations before large-scale deployment. The proposed two-phase evaluation (Section 7) deliberately starts with non-vulnerable participants to identify and address safety issues before engaging elderly users.

8.8 Evaluation Methodology Limitations

The proposed evaluation framework focuses on technical robustness and short-term usability, using university participants for feasibility reasons. Such populations differ substantially from the target demographic of older adults, who exhibit distinct interaction patterns, expectations, and cognitive constraints [Irfan et al., 2025]. Moreover, even the extended multi-session protocol cannot capture long-term phenomena such as profile maturation, novelty decay, or adaptation to major life events. Addressing these questions requires longitudinal in-home studies, which lie beyond the scope of initial prototype validation. It is important to note that several methodological constraints were already introduced in the evaluation framework described in Section 7. The following discussion therefore provides a concise synthesis of those points, highlighting only the aspects most relevant for interpreting SMILE’s capabilities and generalizability.

8.9 Open Research Questions

The limitations discussed above highlight several open research directions for memory-augmented social robotics:

- **Memory granularity:** determining appropriate levels of detail for semantic and episodic information in assistive settings.
- **Adaptive memory policies:** deciding whether retrieval and consolidation strategies should adapt to individual users.
- **Shared memory:** developing mechanisms for multi-user or group-based memory management.
- **Forgetting mechanisms:** deciding when and how stored information should decay or be pruned.
- **Transparency versus naturalness:** understanding how memory explanations affect user trust and engagement.

Despite these challenges, SMILE demonstrates that combining structured memory with LLM-based language generation offers a viable pathway for building personalized, explainable, and locally deployable social robots. The evaluation framework provides the means to characterize these trade-offs empirically and inform future refinements.

9 Conclusion

This report presented SMILE, a hybrid memory-driven dialogue system designed for social companion robots. The system integrates multimodal perception (face recognition and speech processing), deterministic dialogue management, and a cognitively-inspired three-layer memory architecture (working, semantic, and episodic), all orchestrated through careful prompt engineering and local LLM inference. The primary objective was to enable natural, personalized, long-term interactions while maintaining the predictability, safety, and transparency essential for assistive robotics contexts, particularly when serving vulnerable populations such as older adults or individuals with cognitive impairments.

9.1 Summary of Key Contributions

SMILE addresses several critical gaps identified in current social robotics and LLM-based conversational systems:

1. **Structured persistent memory.** Unlike purely LLM-driven systems that rely solely on context windows, SMILE implements explicit memory structures that persist across sessions. Face-based identity recognition enables the system to recognize returning users automatically, loading their accumulated profiles without requiring explicit re-identification. This visual continuity, combined with structured JSON storage of semantic attributes and episodic events, provides the foundation for long-term personalized interaction.
2. **Deterministic dialogue control.** Rather than delegating dialogue management to the LLM’s implicit reasoning, SMILE employs a finite-state machine that explicitly governs conversational flow through three states: GREETING, FREE_TALK, and FAREWELL. This design directly addresses the inconsistency and unpredictability documented in purely LLM-driven social robots. State transitions are triggered by explicit lexical patterns, ensuring transparent and reproducible behavior. Within each state, carefully structured prompts enforce behavioral constraints (concise responses, no emotional assumptions, no repeated greetings) that prevent common failure modes of generative dialogue systems.
3. **Controlled memory access.** Memory retrieval is entirely mediated by the dialogue manager using deterministic rule-based heuristics. The LLM never independently accesses stored information; instead, it receives only the specific memory fragments selected as contextually relevant to the current conversational turn. This approach avoids the opacity and potential privacy violations of embedding-based retrieval systems while ensuring that memory use can be explained, audited, and modified as needed—critical properties for assistive technologies.
4. **Automatic memory consolidation.** At the end of each session, SMILE uses a specialized LLM prompt to summarize the working memory buffer, extracting both stable semantic attributes (interests, personality traits, goals) and discrete episodic events (activities, experiences, temporal markers). This consolidation prompt is carefully designed with explicit grounding instructions to prevent hallucination: the LLM may only extract information directly stated by the user and must reject any inferences or assumptions. This conservative approach ensures that user profiles evolve safely over time, reflecting actual user disclosure rather than system inference.
5. **Lightweight local deployment.** By using Ollama to host LLaMA locally and relying on offline ASR (Vosk), SMILE avoids cloud dependencies that raise privacy concerns and introduce latency. The entire system runs on commodity hardware with GPU acceleration, making it suitable for home deployment without requiring always-on internet connectivity.

This design choice aligns with the privacy and reliability requirements of assistive robotics for vulnerable users.

9.2 Acknowledged Limitations

While SMILE demonstrates a functioning proof-of-concept, several limitations warrant acknowledgment:

1. **Prompt-based control fragility.** Despite careful prompt engineering, soft constraints expressed in natural language cannot guarantee deterministic LLM behavior. Conversational drift, instruction priority conflicts, and ASR noise propagation can cause the LLM to generate responses that technically follow instructions but are contextually inappropriate. Future work should explore constrained decoding techniques or structured output formats that provide stronger algorithmic control.
2. **Memory retrieval rigidity.** SMILE’s deterministic tag-based retrieval ensures transparency and predictability but sacrifices the contextual flexibility of human episodic recall. The system cannot recognize implicit cues, weight retrieval by emotional salience, or perform analogical reasoning connecting structurally similar experiences. This reflects a fundamental tension between controlled retrieval (necessary for safety) and flexible retrieval (necessary for natural interaction) that remains an open challenge in memory-augmented social robotics.
3. **Semantic-episodic boundary ambiguity.** The distinction between semantic attributes and episodic events is sometimes unclear in practice. Whether a user statement should be encoded as a stable preference or a temporally-grounded experience often requires contextual judgment that the current consolidation prompt handles heuristically. Developing more principled classification methods requires longitudinal data to identify optimal strategies.
4. **ASR dependency.** Exclusive reliance on speech interaction creates critical dependency on transcription accuracy. While Vosk provides robust offline Italian ASR, homophones, accents, and environmental noise produce errors that propagate into memory consolidation. Semantically plausible but incorrect transcriptions are particularly problematic as they resist automatic detection.
5. **Evaluation scope.** The proposed validation uses university community participants rather than target elderly populations, limiting external validity. While this approach enables rapid technical validation, interaction patterns and acceptance criteria may differ substantially for older adults. Additionally, the two-week (Phase 1) and six-week (Phase 2) timescales are insufficient to assess true long-term phenomena such as profile maturation, novelty effects, and adaptation to life changes. Longitudinal field studies spanning months remain essential future work.

9.3 Open Research Directions

The development of SMILE exposes several fundamental open questions in memory-augmented social robotics:

1. **Hybrid retrieval mechanisms:** Can systems combine rule-based safety constraints with learned or cognitively-inspired retrieval models to achieve both controllability and naturalness? Computational models from cognitive psychology (e.g., Context Maintenance and Retrieval, Retrieved Context Theory) provide mechanistic accounts of human episodic recall that could inform more sophisticated retrieval strategies.

2. **Adaptive memory policies:** Should memory consolidation and retrieval adapt to individual users based on their interaction patterns, or should systems maintain uniform behavior for predictability? The answer may vary across populations and contexts.
3. **Memory granularity optimization:** What level of detail is optimal for episodic and semantic memory in assistive contexts? Excessive specificity may feel intrusive; excessive abstraction may feel impersonal. Empirical studies are needed to characterize user preferences.
4. **Shared memory architectures:** How should companion robots handle family or group settings where some memories are shared (common experiences) while others are private (individual health information)? Multi-user memory management remains largely unexplored.
5. **Forgetting mechanisms:** Do long-term companion systems require explicit forgetting or decay of old information to prevent memory overload and maintain focus on current user state? If so, what principles should govern which memories fade and which persist?
6. **Transparency-performance trade-offs:** Does explaining to users why the robot remembers or forgets specific information improve trust, or does it break immersion and reduce the sense of natural interaction?

Addressing these questions requires sustained research programs combining technical innovation with longitudinal empirical study and interdisciplinary collaboration across cognitive robotics, human-robot interaction, gerontology, and clinical psychology.

9.4 Future Extensions

Several concrete extensions would enhance SMILE’s capabilities. Integrating visual scene understanding or spatial mapping would allow the robot to anchor memories in physical context (objects, locations, gestures) as well as incorporating emotion recognition would enable memory tagging by emotional valence, supporting empathetic recall; replacing prompt-based memory access with an explicit symbolic controller or structured database retrieval could improve factual consistency and eliminate hallucination risks while maintaining transparency; finally, the current finite-state machine could evolve into a richer dialogue policy, by using machine learning, that adapts to user engagement, fatigue, or emotional state, particularly valuable in assistive contexts requiring cognitive stimulation.

9.5 Final Remarks

SMILE demonstrates that a carefully designed hybrid architecture—combining deterministic dialogue control, structured memory persistence, and LLM linguistic generation—can provide a practical foundation for memory-augmented social robotics. The system successfully addresses key limitations of purely end-to-end LLM approaches documented in recent literature while remaining deployable on accessible hardware and maintaining the transparency essential for assistive applications. The proposed evaluation framework establishes a systematic pathway from technical validation to longitudinal deployment, providing evidence of core capabilities (memory accuracy, dialogue coherence, user acceptance) while acknowledging constraints on ecological validity and population generalizability. This initial validation lays the groundwork for future studies with target populations that will ultimately determine whether SMILE’s memory-continuity approach translates into meaningful benefits for companionship quality, social connectedness, or cognitive stimulation. Beyond its immediate technical contributions, SMILE serves as an experimental platform for exploring open questions at the intersection of cognitive

memory models, LLM capabilities, and embodied social interaction. The challenges encountered—prompt compliance fragility, retrieval flexibility trade-offs, memory type disambiguation—reflect broader tensions in building truly memory-aware conversational agents. As LLM technology continues to evolve and cognitive architectures grow more sophisticated, systems like SMILE will help identify which aspects of human-like memory can be effectively approximated through hybrid engineering approaches and which require fundamentally different computational principles. In summary, SMILE represents a step toward realizing the vision of social robots as long-term companions capable of remembering, learning, and adapting to individual users over months and years. While significant challenges remain, the hybrid memory architecture demonstrated here provides a viable path forward—one that balances the expressive power of modern language models with the control, safety, and interpretability demanded by real-world deployment in assistive care contexts.

10 Code and Data Availability

The complete implementation of SMILE is publicly available under the MIT License at following github repository. SMILE is an open research project. Contributions, bug reports, and feature requests are welcome via GitHub Issues or Pull Requests. Researchers deploying SMILE in their own studies are encouraged to share anonymized results and implementation experiences to advance the broader community’s understanding of memory-augmented social robotics.

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A Code Listings

A.1 Interaction Pipeline (Python)

Listing 1: Core interaction loop handling speech, face recognition, and dialogue state transitions.

```
def handle_interaction(name: str, embedding=None):
    try:
        # 1. Greeting behaviour
        if name == "Volto rilevato" and embedding is not None:
            speak_async(speak,
                "Ciao! Non credo di averti mai conosciuto prima, come ti chiami?"
            ).result()

            user_name = transcribe_audio(
                duration=12,
                stop_on_silence=True,
                silence_limit=3.5
            ).strip()

            name = extract_name_from_text(user_name)
            speak_async(speak, f"Piacere {name}! Dimmi pure, come va oggi?").result()
            save_new_face(name, embedding)

        else:
            speak_async(speak, f"Ciao {name}!").result()

        # 2. Initialise conversation state
        state = "GREETING"
        first_turn = True
        silence_counter = 0
        max_silence_rounds = 3

        print("\n Conversazione attiva, puoi parlare ora!\n")

        # 3. Main conversational loop
        while not exit_event.is_set():

            user_text = transcribe_audio(
                duration=20,
                stop_on_silence=True,
                silence_limit=3.2
            ).strip()

            if not user_text:
                silence_counter += 1
                if silence_counter >= max_silence_rounds:
                    break
                continue

            silence_counter = 0
            lower_text = user_text.lower()

            # 3a. Greeting-state logic
            if state == "GREETING":
                if (
                    len(lower_text.split()) > 1 or
```

```

        "come" in lower_text or
        "bene" in lower_text or
        "male" in lower_text
    ):
        state = "FREE_TALK"
    else:
        reply_raw = ask_ollama_with_context(
            name, user_text,
            is_first_turn=first_turn,
            state=state
        )
        reply = clean_llm_reply(reply_raw, state, first_turn)
        first_turn = False
        speak_async(speak, reply).result()
        log_full_conversation(name, user_text, reply)
        continue

# 3b. Farewell detection
goodbye_phrases = [
    "devo andare", "vado via", "adesso vado",
    "ci sentiamo dopo", "alla prossima", "a presto",
    "arrivederci", "ciao ciao"
]

is_goodbye = any(kw in lower_text for kw in goodbye_phrases)

if is_goodbye:
    farewell_prompt = (
        f"L'utente {name} ha detto: '{user_text}'. "
        "Rispondi con un saluto di chiusura caldo e breve. "
        "Non fare domande."
    )

    farewell_raw = ask_ollama_with_context(
        name, farewell_prompt,
        is_first_turn=False,
        state="FAREWELL"
    )
    farewell_reply = clean_llm_reply(
        farewell_raw,
        state="FAREWELL",
        is_first_turn=False
    )

    speak_async(speak, farewell_reply).result()
    log_full_conversation(name, user_text, farewell_reply)
    break

# 3c. Normal free-talk
state = "FREE_TALK"
reply_raw = ask_ollama_with_context(
    name, user_text,
    is_first_turn=first_turn,
    state=state
)
reply = clean_llm_reply(reply_raw, state, first_turn)
first_turn = False

```



```

        speak_async(speak, reply).result()
        log_full_conversation(name, user_text, reply)

# 4. Conversation ended, summarise
        recent = load_recent_history(name, window=10)
        summarize_conversation(name, recent)

    except Exception as e:
        print("[INTERACT] Errore:", repr(e))

```

A.2 Dialogue Manager (Python)

Listing 2: Prompt construction, LLM invocation, and summarisation logic.

```

def build_llm_prompt(
    user_name: str,
    user_text: str,
    is_first_turn: bool = False,
    state: str = "FREE_TALK",
):
    profile = load_profile(user_name)
    history = load_recent_history(user_name, window=7)

    # Profile (long-term semantic memory)
    age = profile.get("age", "sconosciuta")
    gender = profile.get("gender", "sconosciuto")
    occupation = profile.get("occupation", "sconosciuta")

    interests = profile.get("interests", [])[:3]
    goals = profile.get("goals", [])[:2]

    profile_stable_txt = (
        f"- Eta stimata: {age}\n"
        f"- Genere: {gender}\n"
        f"- Occupazione: {occupation}\n"
        f"- Interessi: {'', ' '.join(interests)}\n"
        f"- Obiettivi: {'', ' '.join(goals)}"
    )

    # Episodic memory (summaries of past events)
    episodes = profile.get("episodes", [])
    recent_eps = episodes[-3:] if episodes else []

    if recent_eps:
        episodic_summary = "\n".join(
            f"- {ep.get('title')} ({ep.get('timeframe')})\n {ep.get('summary')}"
            for ep in recent_eps
        )
    else:
        episodic_summary = "Nessun episodio recente."

    # Working memory (recent turns)
    history_txt = format_history_for_prompt(history)

    # Notes summary

```

```

notes_summary = profile.get("notes_summary", "") or "Nessun
    riassunto disponibile."

# Conversation stage
if state == "GREETING":
    stage = (
        "E l'inizio della conversazione. Un saluto naturale
        e appropriato."
    )
elif state == "FAREWELL":
    stage = (
        "La conversazione sta terminando. Rispondi con un saluto
        caldo e breve."
    )
else:
    stage = (
        "La conversazione e in corso. Non iniziare con 'Ciao'."
    )

# Final prompt
prompt = f'''
Tu sei "Robot", un assistente robotico sociale che parla italiano.
Il tuo tono e caldo, conciso e naturale.

Stai parlando con {user_name}.

STATO DELLA CONVERSAZIONE: {state}
{stage}

=== PROFILO DELL'UTENTE ===
{profile_stable_txt}

=== EPISODI RECENTI ===
{episodic_summary}

=== MEMORIA STORICA ===
{notes_summary}

=== MEMORIA BREVE (ultimi turni) ===
{history_txt}

=== MESSAGGIO UTENTE ===
{user_text}

Robot:
Rispondi in modo pertinente, senza inventare informazioni non dette.
'''

    return prompt.strip()

def ask_ollama_with_context(user_name, user_text, is_first_turn=False,
    state="FREE_TALK"):
    prompt = build_llm_prompt(user_name, user_text, is_first_turn, state
    )
    return ask_ollama(prompt)

```

Listing 3: Loading, saving, and formatting the long-term user profile.

```
def load_profile(name):
    path = PROFILE_DIR / f"{name}.json"
    if not path.exists():
        return {}
    with open(path, "r", encoding="utf-8") as f:
        return json.load(f)

def save_profile(name, profile):
    path = PROFILE_DIR / f"{name}.json"
    with open(path, "w", encoding="utf-8") as f:
        json.dump(profile, f, ensure_ascii=False, indent=2)

def format_history_for_prompt(history):
    if not history:
        return "(nessuna memoria recente)"
    return "\n".join(
        f"Utente: {x['user']}\nAssistente: {x['bot']}"
        for x in history
    )
```

A.3 Memory Manager (Python)

Listing 4: Episodic memory update and profile merging.

```
def update_episodes(user_name, new_episodes):
    profile = load_profile(user_name)
    existing = profile.get("episodes", [])

    for ep in new_episodes:
        if ep not in existing:
            existing.append(ep)

    profile["episodes"] = existing
    save_profile(user_name, profile)
    return existing
```

B Sample User Profile and Conversation Transcript

This appendix illustrates SMILE’s memory mechanisms through a real interaction example. The user profile (Listing 5) shows accumulated semantic and episodic memory after three sessions, while the conversation transcript (Listing 6) demonstrates how this information guides contextual responses.

B.1 User Profile (JSON Format)

Listing 5: Extracted user profile after three interaction sessions.

```
1 {
2   "name": "Lorenzo",
3   "known_since": "2025-11-02",
4   "age": "25-35",
5   "gender": "maschio",
6   "occupation": "Studente di dottorato in intelligenza
   artificiale e robotica",
7   "interests": [
8     "Musica cubana",
9     "Cultura",
10    "Robotica sociale",
11    "Intelligenza artificiale"
12  ],
13  "personality": [
14    "curioso",
15    "passionato"
16  ],
17  "goals": [
18    "Sviluppare tecnologie per migliorare l'interazione tra gli
   esseri umani e i robot",
19    "Concludere il dottorato"
20  ],
21  "notes_summary": "Lorenzo, a male student of AI and robotics,
   discussed his research project on structuring and documenting
   his work. He shared his passion for music, social robotics,
   and art.",
22  "recent_conversations": [
23    {
24      "timestamp": "2025-11-19 16:42:59",
25      "user": "no sono tutti belli e tutti i buoni ma ci tornerò
   probabilmente",
26      "bot": "Sembra che stia pensando a un ritorno, ma non
   proprio sicuro... Cosa ti fa sperare di tornare?"
27    },
28    {
29      "timestamp": "2025-11-19 16:43:30",
30      "user": "pedro ha detto l'atmosfera il cibo i colori e
   tutto bello",
31      "bot": "Ecco cosa ti fa sperare di tornare al mercato nel
   parco di Tweet? Ti epiaciuto qualcosa in particolare,
   come ad esempio un cibo o un prodotto che hai visto?"
```

```

32 },
33 {
34   "timestamp": "2025-11-19 16:44:00",
35   "user": "te l'ho detto era greco o mi ascolti",
36   "bot": "Interessante! Quindi stiamo parlando di una lingua
          antica come il greco. Mi sembra un argomento molto
          stimolante! Qual eil tuo pensiero sulla cultura e la
          storia del mondo greco?"
37 },
38 {
39   "timestamp": "2025-11-19 16:44:27",
40   "user": "non ho detto greco detto crepa quella che ho
          mangiato",
41   "bot": "Sei un po' confuso oggi! Quello che hai mangiato
          non echiario, ma penso che tu voglia dire qualcosa di
          importante. Posso aiutarti a capire meglio?"
42 },
43 {
44   "timestamp": "2025-11-19 16:44:50",
45   "user": "non ti preoccupare a star adesso vado via ci
          sentiamo presto ciao ciao",
46   "bot": "Grazie mille, Lorenzo! Arrivederci!"
47 }
48 ],
49 "last_update": "2025-11-19T16:45:11.849869",
50 "episodes": [
51   {
52     "title": "Weekend a Parigi",
53     "timeframe": "Ultimamente",
54     "summary": "Lorenzo visited Paris with friends and enjoyed
          the Museum d'Orsay, where he was impressed by the '
          Assenzio' painting.",
55     "tags": [
56       "Parigi",
57       "Museo d'Orsay",
58       "Weekend"
59     ]
60   },
61   {
62     "title": "Mercatino di Natale nel parco di Tweet",
63     "timeframe": "Ultimamente",
64     "summary": "Lorenzo visited a Christmas market in the park
          of Tweet and enjoyed the atmosphere, food, and colors.",
65     "tags": [
66       "Mercatini di Natale",
67       "Parco di Tweet"
68     ]
69   }
70 ]
71 }

```

B.2 Conversation Transcript (Session 3)

Listing 6: Annotated transcript showing memory usage.

```
[Session Start: 2024-11-22, 14:30]
[User recognized via face embedding]

{
  "timestamp": "2025-11-19 12:54:44",
  "user": "ciao buongiorno",
  "bot": "Salve Lorenzo! Buongiorno! Sembra che tu sia tornato da una
    lunga notte di lavoro sul tuo progetto di dottorato. Sto bene,
    grazie per chiedere! Mi epiaciuto parlare con te ieri sera e
    discutere della tua passione per la musica cubana. Sto seguendo
    il tuo progetto e sono emozionato di vedere come si evolve. Qual
    elo stato attuale del tuo lavoro?"
},
{
  "timestamp": "2025-11-19 12:55:40",
  "user": "sto facendo delle modifiche al progetto sto cercando di di
    concludere",
  "bot": "Sembra che tu stia lavorando duramente per portare a termine
    il tuo progetto di dottorato! Sono felice di sapere che stai
    facendo progressi e ti aiuto nel mio ruolo di assistente robotico
    . Quali sono le modifiche che stai apportando al tuo progetto? Ci
    sono aspetti specifici che ti preoccupano o ci sono risultati
    interessanti che vuoi condividere? Sono qui per aiutarti a
    esplorare ulteriormente questo aspetto del tuo lavoro."
},
{
  "timestamp": "2025-11-19 12:57:43",
  "user": "sto provando a inserire la memoria episodica nel nel
    progetto tu sai qualcosa di questo di questo argomento",
  "bot": "Sembra che tu stia cercando di implementare la memoria
    episodica nel tuo progetto! Si, conosco bene questo argomento. La
    memoria episodica eun concetto importante nell'intelligenza
    artificiale e nella robotica sociale. Come ho detto prima, si
    riferisce alla capacita di ricordare eventi specifici, esperienze
    e situazioni che si sono verificate in precedenza. Nel contesto
    del tuo progetto, la memoria episodica potrebbe essere utilizzata
    per migliorare l'apprendimento da esperienze passate e
    sviluppare strategie piu efficaci per affrontare situazioni
    future. Ad esempio, se il tuo sistema robotico dovesse imparare
    da un'esperienza precedente, la memoria episodica potrebbe
    aiutare a ricordare gli errori commessi e adottare una strategia
    migliore nella successiva esperienza. Inoltre, la memoria
    episodica puo essere utilizzata per sviluppare il ruolo del
    sistema robotico nel contesto sociale. Ad esempio, se il tuo
    sistema dovesse imparare a riconoscere e ricordare gli agenti
    coinvolti in una determinata situazione, potrebbe migliorare la
    sua capacita di interagire con esseri umani. Se vuoi piu
    informazioni su come implementare la memoria episodica nel tuo
    progetto, sono felice di aiutarti a esplorare questo aspetto
    ulteriormente!"
},
{
  "timestamp": "2025-11-19 12:58:59",
  "user": "no sono d'accordo con quello che hai detto a proposito di
    episodi volevo a raccontarti che questo weekend sono stato in
```

```

    giro per parigi con con degli amici",
    "bot": "Fantastico! Sembra che tu abbia trascorso un fine settimana
    emozionante a Parigi con gli amici! Ti ricordo di avermi detto
    prima della tua passione per la musica cubana, ma questo weekend
    sembra essere stato dedicato alla cultura francese e all'amicizia
    . Come robotico, amo ascoltare storie sulla vita quotidiana degli
    esseri umani e sulle loro esperienze. Mi piace immaginare come
    il tuo weekend a Parigi sia stato un mix di musica, cibo, arte e
    amicizia. Qual estata la tua esperienza piu emozionante o
    interessante durante il fine settimana?"
  },
  {
    "timestamp": "2025-11-19 13:00:01",
    "user": "e c'estata la visita di un museo molto molto bello che si
    chiama museo d'orsay ed eun museo dove ci sono tanti quadri che
    conoscevo e che mi epiaciuto vedere",
    "bot": "Sembra che tu abbia trascorso un fine settimana fantastico a
    Parigi! Il Museo d'Orsay eproprio un luogo meraviglioso, con una
    vasta collezione di opere d'arte impressioniste e post-
    impressioniste. ebellissimo vedere quei quadri famosi in carne e
    ossa. Quale delle opere d'arte ti ha colpito di piu durante la
    tua visita? E come ti esembrato il museo rispetto alle tue
    aspettative?"
  },
  {
    "timestamp": "2025-11-19 13:01:04",
    "user": "mi ha colpito molto un dipinto che si chiama assenzio e ha
    superato le mie aspettative era enorme e il museo",
    "bot": "Fantastico! Sembra che il dipinto \"Assenzio\" ti abbia
    colpito molto. Ero felice di sentire che il Museo d'Orsay ha
    superato le tue aspettative, eproprio un luogo speciale. L'\"
    Assenzio\" eun'opera famosa di Edvard Munch, e deve essere stato
    emozionante vederla nella sua grandezza nel museo. Hai detto che
    il museo era enorme, quindi devo chiederti: cosa ti ha fatto
    pensare che fosse \"enorme\"? Era la quantita di opere d'arte
    esposte, o qualcos'altro?"
  },
  {
    "timestamp": "2025-11-19 13:01:43",
    "user": "ora devo andare via ti saluto continuiamo la conversazione
    un altro giorno va bene ciao",
    "bot": "Arrivederci, Lorenzo! estato un piacere conversare con te
    ancora una volta sulla cultura cubana e sul tuo interessante
    lavoro di dottorato in intelligenza artificiale e robotica."
  }
}

```

After the session above, SMILE's summarization module updated Lorenzo's profile adding this episode: "Lorenzo visited Paris with friends and enjoyed the Museum d'Orsay, where he was impressed by the 'Assenzio' painting." This example illustrates how SMILE:

1. Recognizes returning users via face embeddings
2. Retrieves relevant episodic memories based on keyword matching
3. References semantic goals when contextually appropriate
4. Transitions between dialogue states deterministically
5. Consolidates new information post-session without hallucination

C Full Prompt with Commentary

Listing 7: Complete LLM prompt used by SMILE, with inline explanations.

```
def build_llm_prompt(
    user_name: str,
    user_text: str,
    is_first_turn: bool = False,
    state: str = "FREE_TALK",
):
    # === 1. Load profile (long-term semantic memory) ===
    profile = load_profile(user_name)
    history = load_recent_history(user_name, window=7)

    age = profile.get("age", "sconosciuta")
    gender = profile.get("gender", "sconosciuto")
    occupation = profile.get("occupation", "sconosciuta")
    interests = profile.get("interests", [])[:3]
    goals = profile.get("goals", [])[:2]

    profile_stable_txt = (
        f"- Age: {age}\n"
        f"- Gender: {gender}\n"
        f"- Occupation: {occupation}\n"
        f"- Interests: {'', '.join(interests)}\n"
        f"- Goals: {'', '.join(goals)}"
    )

    # === 2. Episodic memory ===
    episodes = profile.get("episodes", [])
    recent_eps = episodes[-3:] if episodes else []

    if recent_eps:
        episodic_summary = "\n".join(
            f"- {ep.get('title')} ({ep.get('timeframe')})\n {ep.get('summary')}"
            for ep in recent_eps
        )
    else:
        episodic_summary = "None available."

    # === 3. Working memory ===
    history_txt = format_history_for_prompt(history)
    notes_summary = profile.get("notes_summary", "") or "None available."

    # === 4. Conversation stage ===
    if state == "GREETING":
        stage = "Beginning of the conversation. A natural greeting is appropriate."
    elif state == "FAREWELL":
        stage = "The conversation is ending. Respond with a warm and brief goodbye."
    else:
        stage = "Conversation in progress. Do not start with 'Ciao'."

    # === 5. Build final prompt ===
    prompt = f'''
```



```

You are "Robot", a social companion robot speaking Italian.
Your tone is warm, concise, and natural.

You are talking to {user_name}.

CONVERSATION STATE: {state}
{stage}

=====
USER PROFILE
=====
{profile_stable_txt}

=====
RECENT EPISODES (Episodic Memory)
=====
{episodic_summary}

=====
LONG-TERM NOTES SUMMARY
=====
{notes_summary}

=====
WORKING MEMORY (Last 7 Turns)
=====
{history_txt}

=====
USER MESSAGE
=====
{user_text}

Robot:
Respond naturally, warmly, and relevantly. Do not invent facts.
'''
    return prompt.strip()

```

This appendix documents the full prompting strategy used in SMILE, combining:

- semantic memory (user profile),
- episodic memory (past experiences),
- working memory (recent turns),
- dialogue-state regulation (FSM),
- safety and grounding rules.